DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

YEAR I SEMESTER II ORDINARY UNIVERSITY EXAMINATION FOR THE BACHEROR OF SCIENCE IN INFORMATION TECHNOLOGY AND BACHEROR OF SCIENCE IN BUSINESS INFORMATION TECHNOLOGY

CIT 1202: OBJECT ORIENTED PROGRAMMING I

DATE: 11TH DECEMBER 2019 **TIME: 0830 – 1030 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) Objects are key to understanding object-oriented technology as they are used to model the real-world objects that we find in everyday life.

i) Using an example of your choice, explain the two characteristics of an object. (4mks)

ii) Identify and briefly explain any two benefits for bundling code into individual software objects. (4mks)

b) When a class implements an interface, it promises to provide the behavior published by that interface.

i) Briefly describe the term interface as applied in object oriented programming. (1mk)

ii) Using a bicycle class example, write a program that can be used to specify and implement an interface. (8mks)

c) Write a program that searches through an array of 10 integers for a specific value and displays it on the screen or displays the statements "Value not in the array" if the value is not found. (5mks)

d) The Java programming language defines various kinds of variables. Briefly explain how the following variables are used in a java program. (4mks)

i) Instance variables
ii) Class variables

e) Explain any four operations provided by methods in the *java.util.Arrays* class. (4mks)

Question 2: (20 marks)

a) Write a program that declares an int named month whose value represents a month. The code should then display the name of the month, based on the value of month, using the switch statement. (8mks)

b) Write a program that creates an array of nine integer values and then prints each value to a standard output with the formatting shown below. (8mks)

Element at index 0: ---
Element at index 1: ---
.
.
.

Element at index 9: ---

c) The *for statement* ahas another form designed for iteration through Collections and arrays. This is referred to as the *enhanced for* statement. Write a program to demonstrate the use of an *enhanced for* statement. (4mks)

Question 3: (20 marks)

a) A class is a prototype or a blue print for related methods. Outline at least four components that can be included in a class declaration. (4mks)

b) Using an example of your choice, demonstrate how an object can be declared in a java program. (2mks)

c) The Java programming language supports overloading methods, and Java can distinguish between methods with different method signatures.

i) How are overloaded methods differentiated in a java program? (1mk)

ii) Use a program snippet to demonstrate the concept of method overloading. (6mks)

d) Primitive arguments, such as an *int* or a *double*, are passed into methods by *value*. This means that any changes to the values of the parameters exist only within the scope of the method. Write a program to demonstrate this. (5mks)

e) Differentiate between parameters and arguments as used in methods and constructors. (2mks)

Question 4: (20 marks)

a) A class contains constructors that are invoked to create objects from the class blueprint.

i) What is the difference between a constructor declaration and a method declaration? (2mks)

ii) Write a program snippet to demonstrate how a constructor can be declared. (4mks)

b) A typical Java program creates many objects, which interact by invoking methods. Write a program that can be used to creates three objects i.e one Point object and two Rectangle objects. (10mks)

c) The Grid Layout manager lays out GUI components in a rectangular grid. Its general syntax is *GridLayout(r, c, hgap, vgap).* Outline the four positional constraints used in Border layout managers. (4mks)